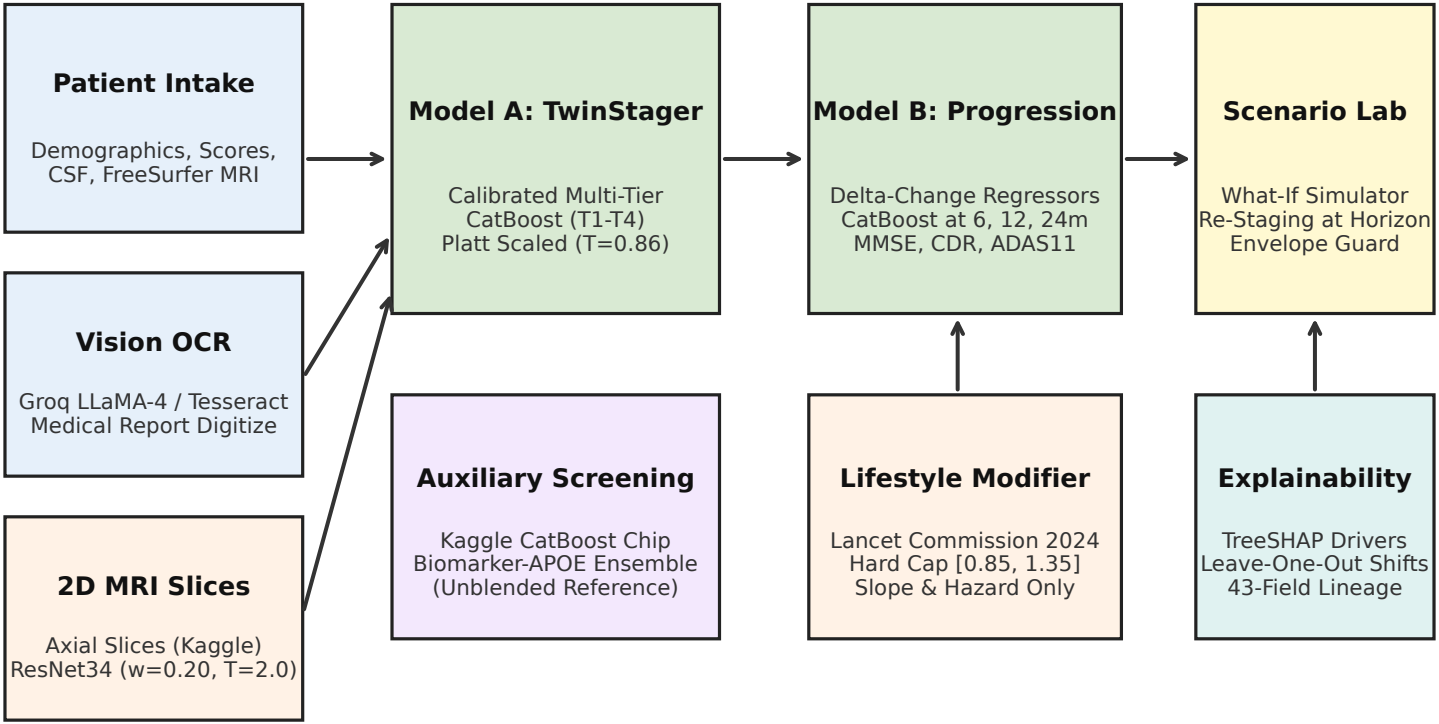System Architecture: End-to-End Multimodal Longitudinal Digital Twin